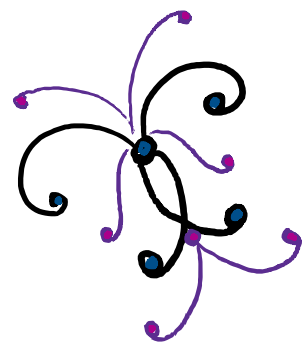


• CORRECTNESS

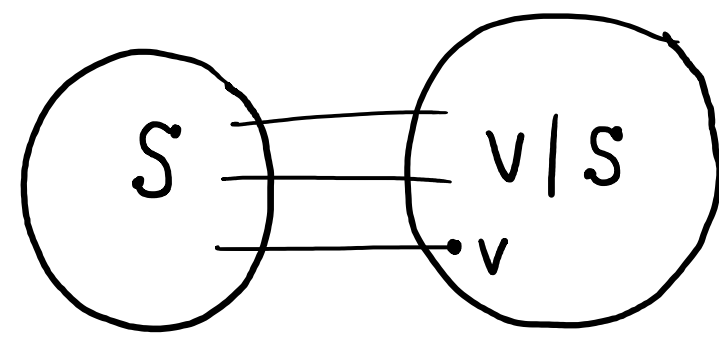
Lemma: Consider the set Visited at an arbitrary point of time in the algorithm's execution. For all $u \in \text{Visited}$, $d(u)$ is the shortest distance from $s \rightsquigarrow u$.



PROOF:
Proof by induction on the size of the set Visited (S).

Base Case: $|S| = 1$
 $S = \{s\}$, $d(s) = 0$

Induction hypothesis: The statement of the lemma is true
 S s.t. $|S| = k$ where $k \geq 1$. $k \leq n-1$



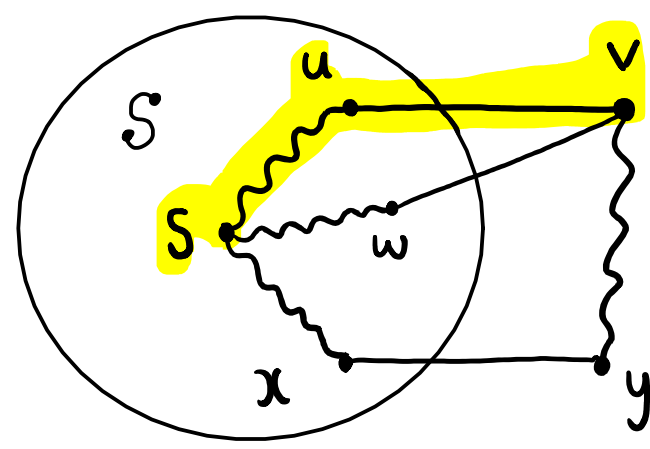
$\frac{N(S) \cap (V \setminus S)}{\text{For each } v \in \dots}$
We compute d' values.
Pick an element that attains min d' value.

$$d' = \min_{(u,v) \in E} \{d(u) + \text{wt}(u,v)\}$$

Through this process, say v attains the min d' value.
Here, $d(v)$ is set to $d'(v)$.

Q: Does there exist any other path than $s \rightsquigarrow u - v$ that gives a shorter $s \rightsquigarrow v$ distance?

$$s \rightsquigarrow w - v \geq d'(v)$$



Let $x \in S$ and $(x,y) \in E$

$$d'(v) \leq d'(y)$$

$$y \neq v \mid \forall e \in E, \text{wt}(e) > 0$$

$$d'(y) = \min_{\substack{(w,y) \in E \\ w \in S}} \{d(w) + \text{wt}(w,y)\}$$

For the sake of contradiction, we assume there exists a shorter path.

Say v attains min d' value through x .

$$\frac{s \rightsquigarrow u - v}{d(s,u) + \text{wt}(u,v)} = d(u) + \text{wt}(u,v)$$

$$\begin{aligned} \frac{s \rightsquigarrow x - y \rightsquigarrow u}{d(x) + \text{wt}(x,y) + l(y,v)} &\geq d'(y) + l(y,v) \\ &\geq d'(v) + l(y,v) \\ &> d'(v) \quad (\text{since } l(y,v) > 0) \end{aligned}$$

$$s \rightsquigarrow w - v \geq d'(v)$$

$$d'(v) = \min_{\substack{(a,v) \in E \\ a \in S \\ w, x \in S}} \begin{cases} d(a) + \text{wt}(a,v) \\ d(w) + \text{wt}(w,v) \\ d(x) + \text{wt}(x,v) \end{cases}$$

$$d'(w) \geq d'(v)$$

If multiple min d' values, break the tie arbitrarily.

From these arguments, we get that $d(v)$ thus computed is the shortest $s \rightsquigarrow v$ distance.

• RUNNING TIME ANALYSIS:

1. Should d' be computed every time?

In each iteration of the while loop, only the neighbors of the min. vertex could have their values updated.

Extract min. from the data structure which stores d' values.

If v attains min value, then $\deg(v)$ many updates are performed.

$(n-1)$ iterations of while loop $\begin{cases} \rightarrow 1 \text{ extract min. operation} \\ \rightarrow \deg(v) \text{ updates} \end{cases}$
 $O(n)$ book keeping

$$\Rightarrow O(n) + (n-1) \text{ extract mins} + 2m \text{ updates}$$

• MINIMUM SPANNING TREES

$$G = (V, E)$$

Spanning Tree: A subgraph of G such that:

- it is a tree. connected, no circuits
- it contains all vertices.

If G is connected, then so must the spanning tree be.

→ Minimum Spanning Tree given $\text{wt}: E \rightarrow \mathbb{R}$
 $\text{wt}(\text{spanning tree } T) = \sum_{e \in T} \text{wt}(e).$

Remark: It is possible to have multiple MSTs.

Algorithmic Task: Find a minimum spanning tree.

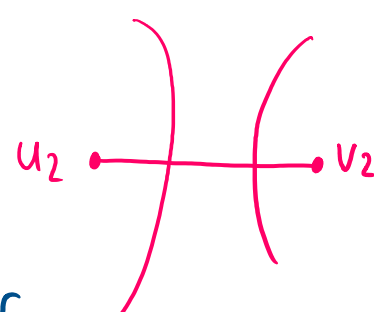
• If the edge weights are distinct, then we have a unique MST.

For the sake of contradiction,

let us assume that T_1 and T_2 both attain min. wt. of C^* .
└─ spanning trees

Suppose $e_1 \in T_1$ but $\notin T_2$
 $e_2 \in T_2$ but $\notin T_1$
 (u_1, v_1)
 (u_2, v_2)

Say $\text{wt}(e_1) < \text{wt}(e_2)$
Remove e_2 from T_2
Add e_1 to $T_2 \setminus \{e_2\}$



NOT PRECISE

"Swapping Arguments"
"Exchange Arguments"

(contd. in next lecture)